

Disassembly Project Report: Stapler

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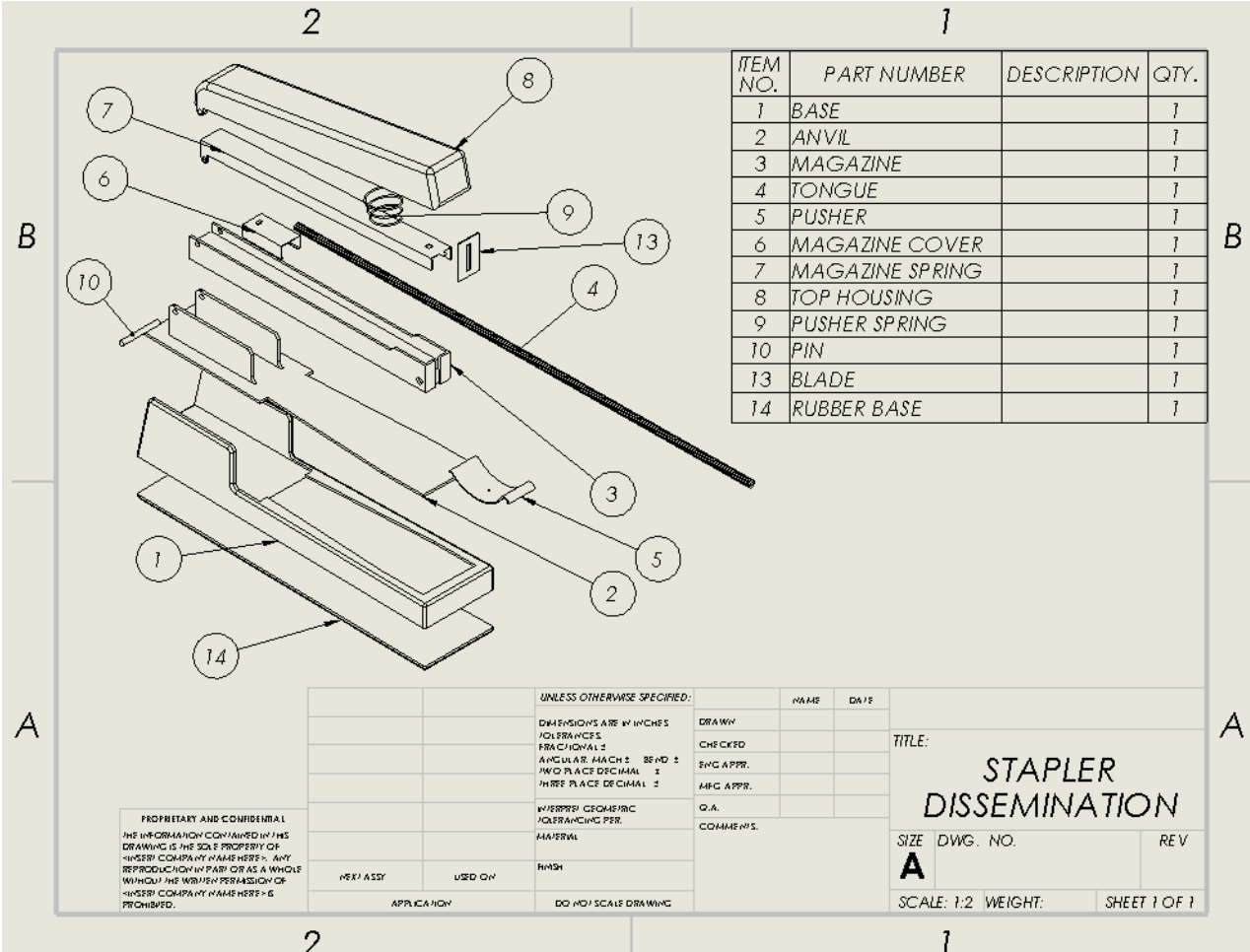
Introduction:

An ordinary stapler goes through a notorious process of design and analysis to be manufactured and sold to consumers, so the product can provide optimal functionality for mass consumption and improve convenience in day-to-day life. As engineers, considerations such as material, design, stress loading, labor, production cost, and revenue are all significant for making any consumer product. Most importantly, engineers must consider who their consumers are and how the product will impact their everyday lives. Staplers are a communal product for consumers such as children, students, office workers, etc. With a plethora of various consumers, creating a universal design while modifying aspects of the products without ruining the quality of the product is ideal. For this report, the design analysis will be catered to office workers as the stapler is essential in their work of writing and exchanging documents/reports. There are approximately over 700,000 office workers in the U.S. currently, so the stapler must have a high production rate to keep up with the supply and demand. With a high production rate comes a steep manufacturing cost, which affects the choice of design and materials. The design specifications to consider are how big or small the head of the stapler is to provide comfort for the consumer, how long the stapler should be to fit different-sized papers, how big the gap is to fit various thicknesses of papers, and what size staple is used. On the other hand, material is also crucial to consider, as the staplers must be able to puncture and fasten all the papers, endure countless cycles of use, and be durable to resist cracking or failure with immense force acting on them. Overall, the analytical approach to developing a successful consumer product requires extensive research and numerous qualifications for the product to provide ideal long-term service to the consumer.

Functionality:

The stapler is comprised of various parts that act in accordance to create a mechanical system that fastens paper together with just a vertical force from a palm. The outer components of the stapler that house the mechanical system are the head and the base. With a standard stapler, the head and the base are designed to be long and rectangular to accommodate the size of the person's palm and to be conveniently stowed on an office desk. The head has an accompanying metal holder called the magazine and a metal pusher. The magazine stores varying amounts of staples, and the pusher aligns the staples to the front of the magazine for use. The pusher is significant to the mechanical system because its motion relies on the head's position. When reloading the stapler, the head is up, and the pusher moves away from the front of the magazine with the stretch of the spring connecting the head to the pusher. With the absence of force on the staples due to the pusher, the person can reload the stapler safely, increasing the factor of safety. Once the head of the stapler is down, the spring compresses and moves towards the front of the magazine. The spring is always under tension to return to its original length, so the pusher drives a constant force on the staples. This allows the stapler to be in a constant state of readiness to be used until running out of staples. Additionally, a metal anvil is attached to the front of the base. The anvil provides the blueprint for bending the staple to fasten the papers. If the anvil wasn't there, the staple would bend randomly or snap under force. The mechanical system is activated with the motion of the head moving towards the base. So, the pin between the head and base must provide a wide range of motion, and the design of the head and base has to align precisely for the magazine and anvil to be in contact. When the magazine comes into contact with the paper, the ram or driver blade pushes a staple out of the magazine. The force on the driver blade allows the staple to puncture the paper, although the staple has blunt ends. Then the staple reaches the anvil, where it then bends to create a flush, secure fastening for the papers. Functionality criteria to consider are whether the parts work cohesively with each other, the amount of force needed to start the mechanical system, how much force the stapler can endure, and the longevity of the parts to be able to produce consistent results after multiple uses.

Disassembly:



¹ Figure above the exploded assembly of a Stapler

Part Type	Description	Price	Quantity	Producer	Manufacturing Process
Standard Part	Pin	\$4.00	1	McMaster Carr	CNC Machining
Standard Part	Pusher Spring	\$6.00	1	McMaster Carr	CNC Wire Coiling
Standard Part	Magazine Spring	\$1.64	1	McMaster Carr	CNC Wire Coiling
Special Purpose Part	Pusher	Approx. \$0.05	1	Sheet Metal Fabrication Plant	Progressive Dye Stamping
Special Purpose Part	Magazine	Approx. \$0.35	1	Sheet Metal Fabrication Plant	Progressive Dye Stamping
Special Purpose Part	Magazine Cover	Approx \$0.10		Sheet Metal Fabrication Plant	Progressive Dye Stamping
Special Purpose Part	Driver Blade	Approx. \$0.10	1	Sheet Metal Fabrication Plant	Progressive Dye Stamping
Special Purpose Part	Anvil	Approx. \$0.12	1	Sheet Metal Fabrication Plant	Progressive Dye Stamping
Special Purpose Part	Top Housing	Approx. \$0.55	1	Plastics Molding Plant	Plastic Injection Molding

Special Purpose Part	Base	Approx. \$0.10	1	Plastics Molding Plant	Plastic Injection Molding
Special Purpose Part	Rubber Base	Approx. \$0.45	1	Molded Rubber Facilities	Rubber Injection Molding
Special Purpose Part	Tongue	Approx. \$0.08	1	Machine Shop	CNC Machining

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Manufacturing Process:

The manufacturing process of a standard stapler relies on high-volume production techniques to ensure durability, precision, and cost-efficiency. The external housing components, specifically the head case and the base, are typically produced through plastic injection molding. This method allows manufacturers to create lightweight, ergonomic, and complex geometries at a low per-unit cost, which is essential for a product with a steep manufacturing rate. The internal mechanical components that endure the highest levels of friction, such as the magazine, pusher, driver, blade (ram), and anvil, are manufactured using progressive die stamping. In this process, sheet steel is fed through a series of specialized dies that punch, coin, and bend the metal into rigid, exact shapes. Finally, the tension springs that connect the head to the pusher and keep the system in a constant state of readiness are fabricated using CNC wire coiling machines, ensuring the springs maintain their structural integrity over countless cycles of use.

Proposed Improvements:

While the current stapler design works well, some design changes could make it more comfortable and reliable to use:

² Parts with “Approx.” listed in price only take into account material cost, as the parts are specially made and are not found for wholesale.

- Thermoplastic elastomer (TPE) over-mold: reduces hand fatigue by creating a soft-grip surface on the head-case, allowing the force from the palm to be distributed more comfortably;
- Anti-jamming design: tightens the space inside the driver blade channel so staples don't buckle or get misaligned when pushing down too hard or off-center;
- Low-staple indicator: adds a small viewing window to the magazine or a bright color to the pusher, so you can easily see when staples are running out.

Conclusion / Findings:

Through the disassembly and analysis of a standard stapler, it was determined that the product is a simple yet highly effective mechanical system designed for durability, reliability, and low-cost mass production. The stapler consists of a combination of standard parts, such as springs and pivot pins, and special-purpose parts, including the magazine, pusher, driver blade, anvil, head case, and base. Each component serves a specific function and works together to convert a user's applied force into the fastening of paper.

The disassembly process revealed that manufacturers carefully select materials and manufacturing methods based on the functional requirements of each component. Plastic injection molding is used for the external housing because it is lightweight, economical, and capable of producing ergonomic shapes, while metal stamping provides the strength and precision required for the internal mechanisms. The use of springs allows the stapler to maintain a constant state of readiness while ensuring consistent performance over many operating cycles.

Overall, the stapler demonstrates how engineering design balances functionality, manufacturability, cost, and user convenience. Although the current design successfully meets the needs of office workers and other consumers, improvements such as a soft-grip handle, anti-jamming features, and a low-staple indicator could further enhance usability and customer satisfaction. This project provided valuable insight into product design, manufacturing processes, material selection, and the importance of designing consumer products that are both efficient and practical.